INFO8006 Introduction to Artificial Intelligence

Exercise session 5: Reasoning over time

Recap

Markov models

The world is a discrete series of time slices, where $\mathbf{X}_t$ denotes the set of *unobservable* state variables and $\mathbf{E}_t$ the set of *observable* evidence variables at time t. We specify a prior $\mathbf{P}(\mathbf{X}_0)$, a *transition model* $\mathbf{P}(\mathbf{X}_t \mid \mathbf{X}_{0:t-1})$ and a *sensor model* $\mathbf{P}(\mathbf{E}_t \mid \mathbf{X}_{0:t}, \mathbf{E}_{0:t-1})$, under

- the first-order *Markov assumption* $\mathbf{P}(\mathbf{X}_t \mid \mathbf{X}_{0:t-1}) = \mathbf{P}(\mathbf{X}_t \mid \mathbf{X}_{t-1})$, which defines a first-order *Markov process*;
- the *sensor Markov assumption* $\mathbf{P}(\mathbf{E}_t \mid \mathbf{X}_{0:t}, \mathbf{E}_{0:t-1}) = \mathbf{P}(\mathbf{E}_t \mid \mathbf{X}_t)$;
- the *stationarity assumption*: the transition and sensor models are the same for all t.

The joint distribution of all variables up to t is then

$$\mathbf{P}(\mathbf{X}_{0:t}, \mathbf{E}_{1:t}) = \mathbf{P}(\mathbf{X}_0) \prod_{i=1}^{t} \mathbf{P}(\mathbf{X}_i \mid \mathbf{X}_{i-1}) \mathbf{P}(\mathbf{E}_i \mid \mathbf{X}_i).$$

Unrolled over time as a Bayesian network, the same slice of variables and arrows repeated at every t, the model is a *dynamic Bayesian network*; a slice may hold several state and evidence variables.

Hidden Markov models

A *hidden Markov model* (HMM) is a Markov process in which the state $\mathbf{X}_t = X_t$ and the evidence $\mathbf{E}_t = E_t$ are both single discrete random variables, with domains $D_{X_t} = \{1, \ldots, S\}$ and $D_{E_t} = \{1, \ldots, R\}$. Its models become matrices:

- the prior $\mathbf{P}(X_0)$ becomes a column vector $\mathbf{f}_0 \in \mathbb{R}_+^S$;
- the transition model $\mathbf{P}(X_t \mid X_{t-1})$ becomes an $S \times S$ *transition matrix* $\mathbf{T}$ such that $\mathbf{T}_{ij} = P(X_t = j \mid X_{t-1} = i)$;
- the sensor model $\mathbf{P}(E_t \mid X_t)$ is an $S \times R$ *sensor matrix* $\mathbf{B}$ such that $\mathbf{B}_{ij} = P(E_t = j \mid X_t = i)$;
- the *observation matrix* $\mathbf{O}_t$ is the diagonal matrix of the column e_t of $\mathbf{B}$, $(\mathbf{O}_t)_{jj} = \mathbf{B}_{je_t}$.

Several evidence variables, conditionally independent given the state, merge into one evidence variable whose values are their combinations, with probabilities the products of theirs. In index notation, $(\mathbf{MN})_{ij} = \sum_k \mathbf{M}_{ik}\mathbf{N}_{kj}$ and $(\mathbf{T}^T\mathbf{f})_j = \sum_i \mathbf{T}_{ij}\mathbf{f}_i$.

Inference tasks

Given the evidence $\mathbf{e}_{1:t}$ collected up to time t, we define

- *filtering*: $\mathbf{P}(\mathbf{X}_t \mid \mathbf{e}_{1:t})$, the *belief state* about the current hidden state $\mathbf{X}_t$;
- *prediction*: $\mathbf{P}(\mathbf{X}_{t+k} \mid \mathbf{e}_{1:t})$ for $k > 0$;
- *smoothing*: $\mathbf{P}(\mathbf{X}_k \mid \mathbf{e}_{1:t})$ for $0 \leq k < t$;
- *most likely explanation*: $\arg\max_{\mathbf{x}_{1:t}} P(\mathbf{x}_{1:t} \mid \mathbf{e}_{1:t})$.

Each is computed by a recursion over t, below. The proportionality signs hide a normalization constant, written $\alpha = 1/Z$.

Filtering and prediction

The *Bayes filter* maintains the belief state, or *forward message*, $\mathbf{f}_{1:t} = \mathbf{P}(\mathbf{X}_t \mid \mathbf{e}_{1:t})$, by alternating two steps:

$$\text{predict:} \qquad \mathbf{P}(\mathbf{X}_{t+1} \mid \mathbf{e}_{1:t}) = \sum_{\mathbf{x}_t} \mathbf{P}(\mathbf{X}_{t+1} \mid \mathbf{x}_t) P(\mathbf{x}_t \mid \mathbf{e}_{1:t}),$$

$$\text{update:} \qquad \mathbf{P}(\mathbf{X}_{t+1} \mid \mathbf{e}_{1:t+1}) \propto \mathbf{P}(\mathbf{e}_{t+1} \mid \mathbf{X}_{t+1}) \mathbf{P}(\mathbf{X}_{t+1} \mid \mathbf{e}_{1:t}),$$

with $Z = P(\mathbf{e}_{t+1} \mid \mathbf{e}_{1:t})$. *Prediction* repeats the predict step up to $t + k$, without update. For an HMM, $\mathbf{f}_{1:t+1} = \alpha\, \mathbf{O}_{t+1} \mathbf{T}^T \mathbf{f}_{1:t}$. For continuous variables, sums are replaced with integrals and probability mass functions with densities:

$$p(\mathbf{x}_{t+1} \mid \mathbf{e}_{1:t+1}) \propto p(\mathbf{e}_{t+1} \mid \mathbf{x}_{t+1}) \int p(\mathbf{x}_{t+1} \mid \mathbf{x}_t) p(\mathbf{x}_t \mid \mathbf{e}_{1:t})\, \mathrm{d}\mathbf{x}_t.$$

Stationary distribution (optional)

Without new evidence, the predicted distribution may converge, as k grows, to a *stationary distribution*: a fixed point $\mathbf{P}(\mathbf{X}_{t+1}) = \mathbf{P}(\mathbf{X}_t)$ of the predict step. For an HMM, it is a vector $\mathbf{f}$ such that $\mathbf{f} = \mathbf{T}^T \mathbf{f}$.

Smoothing

Dividing the evidence $\mathbf{e}_{1:t}$ into $\mathbf{e}_{1:k}$ and $\mathbf{e}_{k+1:t}$,

$$\mathbf{P}(\mathbf{X}_k \mid \mathbf{e}_{1:t}) \propto \mathbf{P}(\mathbf{X}_k \mid \mathbf{e}_{1:k}) \mathbf{P}(\mathbf{e}_{k+1:t} \mid \mathbf{X}_k) = \alpha\, \mathbf{f}_{1:k} \times \mathbf{b}_{k+1:t},$$

where $\times$ is a pointwise multiplication of vectors and the *backward message* $\mathbf{b}_{k+1:t} = \mathbf{P}(\mathbf{e}_{k+1:t} \mid \mathbf{X}_k)$ is computed by backwards recursion, $\mathbf{b}_{k+1:t} = \text{backward}(\mathbf{b}_{k+2:t}, \mathbf{e}_{k+1})$. For an HMM, $\mathbf{b}_{k+1:t} = \mathbf{T}\mathbf{O}_{k+1}\mathbf{b}_{k+2:t}$, from the all-one vector $\mathbf{b}_{t+1:t}$ of size S.

Most likely explanation

The *Viterbi algorithm* replaces the forward message with $\mathbf{m}_{1:t} = \max_{\mathbf{x}_{1:t-1}} \mathbf{P}(\mathbf{x}_{1:t-1}, \mathbf{X}_t \mid \mathbf{e}_{1:t})$, the probability of the most likely path to each state, computed from $\mathbf{m}_{1:1} = \mathbf{f}_{1:1}$ as

$$\mathbf{m}_{1:t+1} \propto \mathbf{P}(\mathbf{e}_{t+1} \mid \mathbf{X}_{t+1}) \max_{\mathbf{x}_t} \mathbf{P}(\mathbf{X}_{t+1} \mid \mathbf{x}_t) \mathbf{m}_{1:t}.$$

For each value $\mathbf{x}_{t+1}$, it records the maximizing predecessor $\psi_{t+1}(\mathbf{x}_{t+1}) = \arg\max_{\mathbf{x}_t} P(\mathbf{x}_{t+1} \mid \mathbf{x}_t) \mathbf{m}_{1:t}(\mathbf{x}_t)$. The most likely path ends in $\mathbf{x}_t^* = \arg\max_{\mathbf{x}_t} \mathbf{m}_{1:t}(\mathbf{x}_t)$ and is recovered backwards, $\mathbf{x}_k^* = \psi_{k+1}(\mathbf{x}_{k+1}^*)$.

Cheat sheet for Gaussian distributions

If $\mathbf{x}$ and $\mathbf{y}$ have the joint Gaussian distribution

$$p\begin{pmatrix}\mathbf{x}\\ \mathbf{y}\end{pmatrix} = \mathcal{N}\left(\begin{pmatrix}\mathbf{x}\\ \mathbf{y}\end{pmatrix} \middle| \begin{pmatrix}\mathbf{a}\\ \mathbf{b}\end{pmatrix}, \begin{pmatrix}\mathbf{A} & \mathbf{C}\\ \mathbf{C}^T & \mathbf{B}\end{pmatrix}\right),$$

then the marginal and conditional distributions of $\mathbf{x}$ and $\mathbf{y}$ are given by the following identities, whose letters are generic (they are not the matrices of the HMM or of the Kalman filter):

$$\begin{aligned} p(\mathbf{x}) &= \mathcal{N}(\mathbf{x} \mid \mathbf{a}, \mathbf{A}),\\ p(\mathbf{y}) &= \mathcal{N}(\mathbf{y} \mid \mathbf{b}, \mathbf{B}),\\ p(\mathbf{x} \mid \mathbf{y}) &= \mathcal{N}(\mathbf{x} \mid \mathbf{a} + \mathbf{C}\mathbf{B}^{-1}(\mathbf{y} - \mathbf{b}), \mathbf{A} - \mathbf{C}\mathbf{B}^{-1}\mathbf{C}^T),\\ p(\mathbf{y} \mid \mathbf{x}) &= \mathcal{N}(\mathbf{y} \mid \mathbf{b} + \mathbf{C}^T\mathbf{A}^{-1}(\mathbf{x} - \mathbf{a}), \mathbf{B} - \mathbf{C}^T\mathbf{A}^{-1}\mathbf{C}). \end{aligned}$$

If $p(\mathbf{x}) = \mathcal{N}(\mathbf{x} \mid \boldsymbol{\mu}, \boldsymbol{\Sigma})$ and $p(\mathbf{y} \mid \mathbf{x}) = \mathcal{N}(\mathbf{y} \mid \mathbf{H}\mathbf{x} + \mathbf{u}, \mathbf{R})$, then the joint distribution of $\mathbf{x}$ and $\mathbf{y}$ is Gaussian with

$$p\begin{pmatrix}\mathbf{x}\\ \mathbf{y}\end{pmatrix} = \mathcal{N}\left(\begin{pmatrix}\mathbf{x}\\ \mathbf{y}\end{pmatrix} \middle| \begin{pmatrix}\boldsymbol{\mu}\\ \mathbf{H}\boldsymbol{\mu} + \mathbf{u}\end{pmatrix}, \begin{pmatrix}\boldsymbol{\Sigma} & \boldsymbol{\Sigma}\mathbf{H}^T\\ \mathbf{H}\boldsymbol{\Sigma} & \mathbf{H}\boldsymbol{\Sigma}\mathbf{H}^T + \mathbf{R}\end{pmatrix}\right).$$

Kalman filter

The *Kalman filter* is the Bayes filter for continuous state and evidence variables under a Gaussian prior $p(\mathbf{x}_0)$, a linear Gaussian transition model and a linear Gaussian sensor model,

$$\begin{aligned} p(\mathbf{x}_{t+1} \mid \mathbf{x}_t) &= \mathcal{N}(\mathbf{x}_{t+1} \mid \mathbf{A}\mathbf{x}_t + \mathbf{b}, \mathbf{Q}),\\ p(\mathbf{e}_t \mid \mathbf{x}_t) &= \mathcal{N}(\mathbf{e}_t \mid \mathbf{H}\mathbf{x}_t + \mathbf{d}, \mathbf{R}). \end{aligned}$$

Under these assumptions, the predict and update steps both yield Gaussian distributions, $p(\mathbf{x}_{t+1} \mid \mathbf{e}_{1:t}) = \mathcal{N}(\mathbf{x}_{t+1} \mid \boldsymbol{\mu}^-_{t+1}, \boldsymbol{\Sigma}^-_{t+1})$ and $p(\mathbf{x}_{t+1} \mid \mathbf{e}_{1:t+1}) = \mathcal{N}(\mathbf{x}_{t+1} \mid \boldsymbol{\mu}_{t+1}, \boldsymbol{\Sigma}_{t+1})$, whose parameters are derived in closed form with the Gaussian identities above.

Particle filter

When the models are not linear Gaussian, a *particle filter* approximates $p(\mathbf{x}_t \mid \mathbf{e}_{1:t})$ with N samples $\mathbf{x}^1_t, \ldots, \mathbf{x}^N_t$, the *particles*. Each step propagates every particle through the transition model, $\mathbf{x}^i_{t+1} \sim p(\mathbf{x}_{t+1} \mid \mathbf{x}^i_t)$, weights it by $w^i = p(\mathbf{e}_{t+1} \mid \mathbf{x}^i_{t+1})$, then resamples N particles with probabilities proportional to the weights.

Exercises

In session: exercises 1 and 3 (November 5), 4 and 5 with the Kalman notebook (November 12). *Recommended before the next session:* exercise 2 (d)–(e).

Exercise 1 Load of a shared server (Exam, January 2023)

A scheduling program submits jobs to a shared compute server, whose load varies over time. Each job runs under either a heavy or a light load. From past logs, the program knows that if a job runs under a heavy load, there is a 40% chance that the next job also runs under a heavy load. If a job runs under a light load, there is a 70% chance that the next job also runs under a light load. The program cannot see the load, but it sees the response time of each job, which is either fast, normal or slow. If the load is heavy, the response time is fast with probability 20% and slow with probability 60%. If the load is light, it is fast with probability 50% and slow with probability 30%.

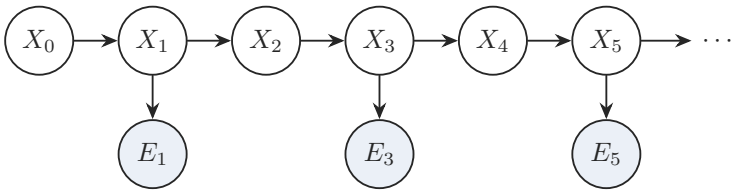
(a) Define the components of a hidden Markov model (HMM), where the state $X_t \in \{\texttt{heavy}, \texttt{light}\}$ is the load under which the t-th job runs and the evidence $E_t \in \{\texttt{fast}, \texttt{normal}, \texttt{slow}\}$ is its response time. Before the first job, the program has no information about the load, so that the prior $\mathbf{P}(X_0)$ is uniform.

Solution. With the states in the order (`heavy`, `light`) and the evidence values in the order (`fast`, `normal`, `slow`), the prior vector, the transition matrix $\mathbf{T}_{ij} = P(X_t = j \mid X_{t-1} = i)$ and the sensor matrix $\mathbf{B}_{ij} = P(E_t = j \mid X_t = i)$ are

$$\mathbf{f}_0 = \begin{pmatrix} 0.5 \\ 0.5 \end{pmatrix}, \quad \mathbf{T} = \begin{pmatrix} 0.4 & 0.6 \\ 0.3 & 0.7 \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} 0.2 & 0.2 & 0.6 \\ 0.5 & 0.2 & 0.3 \end{pmatrix}.$$

(b) At some point, the server stops reporting the response time of every other job (the 2nd, the 4th, ...), so that the program only observes the 1st, 3rd, 5th, ... jobs. Sketch a Bayesian network describing this new process.

Solution.



Keeping E_2, E_4, ... as unobserved (unshaded) nodes is equally valid: they sum out of every query.

(c) Express, in terms of the HMM components, the distribution $\mathbf{P}(X_t \mid e_{1:t:2})$ of the load for the t-th job given the observations $e_{1:t:2}$ of every other job. Note that $e_{1:6:2} = (e_1, e_3, e_5)$ and $e_{1:7:2} = (e_1, e_3, e_5, e_7)$. Separate the cases where t is even and odd.

Solution. Let $\mathbf{f}_{1:t} = \mathbf{P}(X_t \mid e_{1:t:2})$ be the forward message over the evidence observed up to t. Then

$$\mathbf{f}_{1:2n} = \mathbf{T}^T \mathbf{f}_{1:2n-1},$$
$$\mathbf{f}_{1:2n+1} = \alpha\, \mathbf{O}_{2n+1} \mathbf{T}^T \mathbf{f}_{1:2n} :$$

at even t, there is no evidence and only the predict step remains. Both follow from the Bayes filter. For $t = 4$, by the Markov assumption,

$$\mathbf{P}(X_4 \mid e_1, e_3) = \sum_{x_3} \mathbf{P}(X_4 \mid x_3, e_1, e_3) P(x_3 \mid e_1, e_3) = \sum_{x_3} \mathbf{P}(X_4 \mid x_3) P(x_3 \mid e_1, e_3),$$

that is, $\mathbf{f}_{1:4,j} = \sum_i \mathbf{T}_{ij}\mathbf{f}_{1:3,i}$ for every j, or $\mathbf{f}_{1:4} = \mathbf{T}^T\mathbf{f}_{1:3}$. For $t = 3$, by Bayes' rule, then the sensor Markov and the Markov assumptions,

$$\begin{aligned}\mathbf{P}(X_3 \mid e_1, e_3) &= \alpha\,\mathbf{P}(e_3 \mid X_3, e_1)\mathbf{P}(X_3 \mid e_1) \\ &= \alpha\,\mathbf{P}(e_3 \mid X_3)\sum_{x_2}\mathbf{P}(X_3 \mid x_2)P(x_2 \mid e_1),\end{aligned}$$

that is, $\mathbf{f}_{1:3,j} = \alpha\,\mathbf{B}_{je_3}\sum_i \mathbf{T}_{ij}\mathbf{f}_{1:2,i}$ for every j, or $\mathbf{f}_{1:3} = \alpha\,\mathbf{O}_3\mathbf{T}^T\mathbf{f}_{1:2}$, since the observation matrix $\mathbf{O}_3$ is diagonal with entries $\mathbf{B}_{je_3}$.

(d) Given the observations $e_{1:3:2} = (e_1, e_3) = (\texttt{slow}, \texttt{fast})$, compute the probability that the 3rd job runs under a heavy load.

Solution.

$$\mathbf{f}_{1:1} = \alpha\,\mathbf{O}_1\mathbf{T}^T\mathbf{f}_0 = \alpha\begin{pmatrix}0.6 & 0\\ 0 & 0.3\end{pmatrix}\begin{pmatrix}0.4 & 0.3\\ 0.6 & 0.7\end{pmatrix}\begin{pmatrix}0.5\\ 0.5\end{pmatrix} \approx \begin{pmatrix}0.5185\\ 0.4815\end{pmatrix}$$

$$\mathbf{f}_{1:2} = \mathbf{T}^T\mathbf{f}_{1:1} = \begin{pmatrix}0.4 & 0.3\\ 0.6 & 0.7\end{pmatrix}\begin{pmatrix}0.5185\\ 0.4815\end{pmatrix} \approx \begin{pmatrix}0.3519\\ 0.6481\end{pmatrix}$$

$$\mathbf{f}_{1:3} = \alpha\,\mathbf{O}_3\mathbf{T}^T\mathbf{f}_{1:2} = \alpha\begin{pmatrix}0.2 & 0\\ 0 & 0.5\end{pmatrix}\begin{pmatrix}0.4 & 0.3\\ 0.6 & 0.7\end{pmatrix}\begin{pmatrix}0.3519\\ 0.6481\end{pmatrix} \approx \begin{pmatrix}0.1678\\ 0.8322\end{pmatrix}$$

Finally, $P(X_3 = \texttt{heavy} \mid e_{1:3:2}) \approx 0.1678$.

(e) Without any observation, which fraction of the jobs runs under a heavy load in the long run?

Solution. This is the stationary distribution $\mathbf{f} = \mathbf{T}^T\mathbf{f}$. With $\mathbf{f} = \begin{pmatrix}h & 1-h\end{pmatrix}^T$, the first row reads $h = 0.4\,h + 0.3\,(1-h)$, hence $h = 1/3$: in the long run, one job in three runs under a heavy load, whatever the load of the first job.

Exercise 2 Umbrella world (AIMA ch. 14)

A security robot guards a secret underground installation. It would like to know whether it is raining today, but its only access to the outside world occurs each morning, when it sees the director coming in with, or without, an umbrella. For each day t, the evidence is a single variable $\mathit{Umbrella}_t \in \{\text{true}, \text{false}\}$, or U_t for short, *i.e.* whether the umbrella appears or not, and the (hidden) state is a single variable $\mathit{Rain}_t \in \{\text{true}, \text{false}\}$, or R_t, *i.e.* whether it is raining or not.

The robot's model assumes that from one day $t-1$ to the next t, the chances that the weather stays the same are 70 %. It also assumes that the director brings an umbrella 90 % of the time when it is raining, and 20 % of the time otherwise.

(a) The robot represents the umbrella world as a Markov model. What formal assumptions correspond to these beliefs?

Solution. The first belief describes the umbrella world as a *first-order Markov process*,

$$\mathbf{P}(R_t \mid R_{0:t-1}) = \mathbf{P}(R_t \mid R_{t-1}),$$

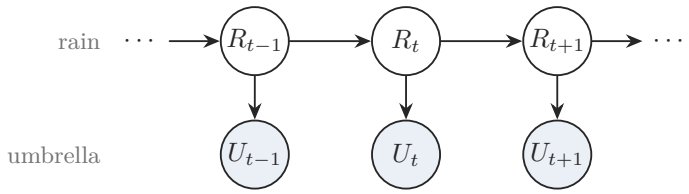
i.e. R_t is independent of $R_{0:t-2}$ given R_{t-1}. The second belief is the *sensor Markov assumption*,

$$\mathbf{P}(U_t \mid R_{0:t}, U_{1:t-1}) = \mathbf{P}(U_t \mid R_t).$$

Both beliefs hold with the same numbers for every day t: this is the *stationarity assumption*.

(b) Sketch a Bayesian network structure describing the umbrella world and provide the transition and sensor models.

Solution.



R_{t-1}	$P(r_t \mid R_{t-1})$
true	0.7
false	0.3

R_t	$P(u_t \mid R_t)$
true	0.9
false	0.2

With the states and the evidence values in the order (true, false), the transition and sensor matrices are

$$\mathbf{T} = \begin{pmatrix} 0.7 & 0.3 \\ 0.3 & 0.7 \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} 0.9 & 0.1 \\ 0.2 & 0.8 \end{pmatrix}.$$

(c) Express the distributions $\mathbf{P}(R_{t+1} \mid R_{t-1})$, $\mathbf{P}(U_t \mid R_{t-1})$ and $\mathbf{P}(R_t \mid R_{t-1}, U_t)$ in terms of the transition and sensor models.

Solution. Summing over r_t, and with the sensor Markov assumption for the second,

$$\mathbf{P}(R_{t+1} \mid R_{t-1}) = \sum_{r_t} \mathbf{P}(R_{t+1} \mid r_t) P(r_t \mid R_{t-1}),$$
$$\mathbf{P}(U_t \mid R_{t-1}) = \sum_{r_t} \mathbf{P}(U_t \mid r_t) P(r_t \mid R_{t-1}).$$

In matrix form, their entries (i, j) are $\sum_k \mathbf{T}_{ik}\mathbf{T}_{kj}$ and $\sum_k \mathbf{T}_{ik}\mathbf{B}_{kj}$:

$$\mathbf{TT} = \begin{pmatrix} 0.58 & 0.42 \\ 0.42 & 0.58 \end{pmatrix}, \quad \mathbf{TB} = \begin{pmatrix} 0.69 & 0.31 \\ 0.41 & 0.59 \end{pmatrix}.$$

By Bayes' rule and the sensor Markov assumption,

$$\mathbf{P}(R_t \mid R_{t-1}, U_t) = \frac{\mathbf{P}(U_t \mid R_t)\mathbf{P}(R_t \mid R_{t-1})}{\mathbf{P}(U_t \mid R_{t-1})},$$

which gives

R_{t-1}	U_t	$P(r_t \mid R_{t-1}, U_t)$
true	true	0.913
true	false	0.226
false	true	0.659
false	false	0.051

For instance, $P(r_t \mid r_{t-1}, u_t) = 0.9 \times 0.7/0.69 \approx 0.913$.

(d) Suppose the robot observes an unending sequence of days on which the umbrella appears. Show that, as the days go by, the probability of rain on the current day tends monotonically towards a fixed point. Calculate this fixed point.

Solution. Let $\pi_t = P(r_t \mid u_{1:t})$, the filtered probability of rain. By the Bayes filter, $\mathbf{f}_{1:t} = \alpha\,\mathbf{O}\mathbf{T}^T\mathbf{f}_{1:t-1}$, with the observation matrix $\mathbf{O} = \mathrm{diag}(0.9, 0.2)$ of an umbrella day, and $\mathbf{T}^T = \mathbf{T}$ (symmetric here):

$$\begin{pmatrix} \pi_t \\ 1-\pi_t \end{pmatrix} = \alpha \begin{pmatrix} 0.9 & 0 \\ 0 & 0.2 \end{pmatrix} \begin{pmatrix} 0.7 & 0.3 \\ 0.3 & 0.7 \end{pmatrix} \begin{pmatrix} \pi_{t-1} \\ 1-\pi_{t-1} \end{pmatrix} = \alpha \begin{pmatrix} 0.27 + 0.36\,\pi_{t-1} \\ 0.14 - 0.08\,\pi_{t-1} \end{pmatrix},$$

hence $\pi_t = g(\pi_{t-1})$ with

$$g(\pi) = \frac{0.27 + 0.36\,\pi}{0.41 + 0.28\,\pi}.$$

A fixed point $\pi^* = g(\pi^*)$ satisfies $0.28\,\pi^{*2} + 0.05\,\pi^* - 0.27 = 0$, whose root in $[0, 1]$ is

$$\pi^* = \frac{-0.05 + \sqrt{0.05^2 + 4 \times 0.27 \times 0.28}}{2 \times 0.28} \approx 0.897.$$

The other root, $\pi^- \approx -1.075$, is negative. Monotonicity follows from two facts on $[0, 1]$.

- g is increasing: $g'(\pi) = \frac{0.36 \times 0.41 - 0.27 \times 0.28}{(0.41 + 0.28\,\pi)^2} = \frac{0.072}{(0.41 + 0.28\,\pi)^2} > 0$.
- $g(\pi) - \pi = \frac{-0.28\,(\pi - \pi^*)(\pi - \pi^-)}{0.41 + 0.28\,\pi}$ has the sign of $\pi^* - \pi$, since $\pi - \pi^- > 0$.

If $\pi_{t-1} < \pi^*$, the second fact gives $\pi_t > \pi_{t-1}$, and the first $\pi_t = g(\pi_{t-1}) < g(\pi^*) = \pi^*$. The sequence is increasing and bounded by π^*, so it converges, to a fixed point of g by continuity, which can only be π^*. The case $\pi_{t-1} > \pi^*$ is symmetric, with a decreasing sequence.

(e) Now consider forecasting further and further into the future, given t umbrella observations. Is there a fixed point? If yes, compute its exact value.

Solution. Let $\pi_k = P(r_{t+k} \mid u_{1:t})$. This is prediction, $\mathbf{P}(R_{t+k} \mid u_{1:t}) = \mathbf{T}^T\mathbf{P}(R_{t+k-1} \mid u_{1:t})$, so that

$$\pi_k = 0.7\,\pi_{k-1} + 0.3\,(1 - \pi_{k-1}) = 0.3 + 0.4\,\pi_{k-1}, \quad \textit{i.e.} \quad \pi_k - 0.5 = 0.4\,(\pi_{k-1} - 0.5).$$

The distance to 0.5 shrinks by a factor 0.4 at each step, and π_k tends to the fixed point $\pi^* = 0.5$, the stationary distribution of the chain: without new evidence, uncertainty about the state accumulates.

Exercise 3 Coin-tossing machine

The surveillance program of a casino watches a coin-tossing machine. The machine holds 3 biased coins a, b and c, whose probabilities of heads are 80 %, 50 % and 20 %, respectively. It first picks one of the coins at random. Then, 4 times, it either keeps the current coin with probability $\frac{2}{3}$ or replaces it by one of the two other coins, chosen at random, then tosses the coin and shows the result. The program only sees the results: heads, heads, tails, heads.

(a) Provide a hidden Markov model (HMM) that describes the process.

Solution. The state $X_t \in \{a, b, c\}$ is the coin tossed at time t, and X_0 the coin picked first. The evidence $E_t \in \{\text{heads}, \text{tails}\}$ is the result of the toss: $e_1 = e_2 = e_4 =$ heads and $e_3 =$ tails. With the states in the order (a, b, c) and the results in the order (heads, tails),

$$\mathbf{f}_0 = \begin{pmatrix} \frac{1}{3} \\ \frac{1}{3} \\ \frac{1}{3} \end{pmatrix}, \quad \mathbf{T} = \begin{pmatrix} \frac{2}{3} & \frac{1}{6} & \frac{1}{6} \\ \frac{1}{6} & \frac{2}{3} & \frac{1}{6} \\ \frac{1}{6} & \frac{1}{6} & \frac{2}{3} \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} 0.8 & 0.2 \\ 0.5 & 0.5 \\ 0.2 & 0.8 \end{pmatrix}.$$

(b) What are the probabilities of the last coin given the sequence of results?

Solution. This is filtering, $\mathbf{f}_{1:4} = \mathbf{P}(X_4 \mid e_{1:4})$. The Bayes filter gives

$$\begin{aligned}\mathbf{P}(X_t \mid e_{1:t}) &= \alpha\, \mathbf{P}(e_t \mid X_t, e_{1:t-1}) \sum_{x_{t-1}} \mathbf{P}(X_t \mid x_{t-1}, e_{1:t-1}) P(x_{t-1} \mid e_{1:t-1}) \\ &= \alpha\, \mathbf{P}(e_t \mid X_t) \sum_{x_{t-1}} \mathbf{P}(X_t \mid x_{t-1}) P(x_{t-1} \mid e_{1:t-1}),\end{aligned}$$

i.e. $\mathbf{f}_{1:t} = \alpha\, \mathbf{O}_t \mathbf{T}^T \mathbf{f}_{1:t-1}$, where the observation matrix $\mathbf{O}_t$ is the diagonal matrix of the column e_t of $\mathbf{B}$:

$$\mathbf{O}_1 = \mathbf{O}_2 = \mathbf{O}_4 = \begin{pmatrix} 0.8 & 0 & 0 \\ 0 & 0.5 & 0 \\ 0 & 0 & 0.2 \end{pmatrix}, \quad \mathbf{O}_3 = \begin{pmatrix} 0.2 & 0 & 0 \\ 0 & 0.5 & 0 \\ 0 & 0 & 0.8 \end{pmatrix}.$$

Then

$$\begin{aligned}\mathbf{f}_{1:1} &= \alpha\, \mathbf{O}_1 \mathbf{T}^T \mathbf{f}_0 = \alpha \begin{pmatrix} 0.8 & 0 & 0 \\ 0 & 0.5 & 0 \\ 0 & 0 & 0.2 \end{pmatrix} \begin{pmatrix} \frac{2}{3} & \frac{1}{6} & \frac{1}{6} \\ \frac{1}{6} & \frac{2}{3} & \frac{1}{6} \\ \frac{1}{6} & \frac{1}{6} & \frac{2}{3} \end{pmatrix} \begin{pmatrix} \frac{1}{3} \\ \frac{1}{3} \\ \frac{1}{3} \end{pmatrix} = \alpha \begin{pmatrix} \frac{4}{15} \\ \frac{1}{6} \\ \frac{1}{15} \end{pmatrix} = \begin{pmatrix} \frac{8}{15} \\ \frac{1}{3} \\ \frac{2}{15} \end{pmatrix}, \\ \mathbf{f}_{1:2} &= \alpha\, \mathbf{O}_2 \mathbf{T}^T \mathbf{f}_{1:1} \approx \begin{pmatrix} 0.619 & 0.298 & 0.083 \end{pmatrix}^T, \\ \mathbf{f}_{1:3} &= \alpha\, \mathbf{O}_3 \mathbf{T}^T \mathbf{f}_{1:2} \approx \begin{pmatrix} 0.227 & 0.376 & 0.397 \end{pmatrix}^T, \\ \mathbf{f}_{1:4} &= \alpha\, \mathbf{O}_4 \mathbf{T}^T \mathbf{f}_{1:3} \approx \begin{pmatrix} 0.472 & 0.374 & 0.154 \end{pmatrix}^T.\end{aligned}$$

(c) What are the probabilities of the coin picked first given the sequence of results? And of the first coin tossed?

Solution. This is smoothing, $\mathbf{P}(X_k \mid e_{1:4})$ for $k = 0$ and $k = 1$:

$$\mathbf{P}(X_k \mid e_{1:t}) = \alpha\, \mathbf{P}(X_k \mid e_{1:k}) \mathbf{P}(e_{k+1:t} \mid X_k) = \alpha\, \mathbf{f}_{1:k} \times \mathbf{b}_{k+1:t},$$

with $\mathbf{f}_{1:0} = \mathbf{f}_0$. The backward message satisfies

$$\begin{aligned}\mathbf{P}(e_{k+1:t} \mid X_k) &= \sum_{x_{k+1}} P(e_{k+1:t} \mid x_{k+1}) \mathbf{P}(x_{k+1} \mid X_k) \\ &= \sum_{x_{k+1}} P(e_{k+1} \mid x_{k+1}) P(e_{k+2:t} \mid x_{k+1}) \mathbf{P}(x_{k+1} \mid X_k),\end{aligned}$$

i.e. $\mathbf{b}_{k+1:t} = \mathbf{T}\mathbf{O}_{k+1}\mathbf{b}_{k+2:t}$, starting from the all-one vector $\mathbf{b}_{5:4} = \begin{pmatrix} 1 & 1 & 1 \end{pmatrix}^T$. Then

$$\begin{aligned}\mathbf{b}_{4:4} &= \mathbf{T}\mathbf{O}_4\mathbf{b}_{5:4} = \begin{pmatrix} \frac{2}{3} & \frac{1}{6} & \frac{1}{6} \\ \frac{1}{6} & \frac{2}{3} & \frac{1}{6} \\ \frac{1}{6} & \frac{1}{6} & \frac{2}{3} \end{pmatrix} \begin{pmatrix} 0.8 & 0 & 0 \\ 0 & 0.5 & 0 \\ 0 & 0 & 0.2 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 0.65 \\ 0.5 \\ 0.35 \end{pmatrix}, \\ \mathbf{b}_{3:4} &= \mathbf{T}\mathbf{O}_3\mathbf{b}_{4:4} = \begin{pmatrix} 0.175 & 0.235 & 0.250 \end{pmatrix}^T, \\ \mathbf{b}_{2:4} &= \mathbf{T}\mathbf{O}_2\mathbf{b}_{3:4} \approx \begin{pmatrix} 0.121 & 0.110 & 0.076 \end{pmatrix}^T, \\ \mathbf{b}_{1:4} &= \mathbf{T}\mathbf{O}_1\mathbf{b}_{2:4} \approx \begin{pmatrix} 0.076 & 0.055 & 0.035 \end{pmatrix}^T.\end{aligned}$$

Eventually,

$$\begin{aligned}\mathbf{P}(X_0 \mid e_{1:4}) &= \alpha\, \mathbf{f}_0 \times \mathbf{b}_{1:4} \approx \begin{pmatrix} 0.457 & 0.331 & 0.212 \end{pmatrix}^T, \\ \mathbf{P}(X_1 \mid e_{1:4}) &= \alpha\, \mathbf{f}_{1:1} \times \mathbf{b}_{2:4} \approx \begin{pmatrix} 0.580 & 0.329 & 0.091 \end{pmatrix}^T.\end{aligned}$$

(d) What is the most likely sequence of tossed coins?

Solution. The Viterbi algorithm computes $\mathbf{m}_{1:t} = \max_{x_{1:t-1}} \mathbf{P}(x_{1:t-1}, X_t \mid e_{1:t})$, from $\mathbf{m}_{1:1} = \mathbf{f}_{1:1}$ and, for each state j,

$$\mathbf{m}_{1:t+1}(j) = \alpha\, \mathbf{B}_{je_{t+1}} \max_i \mathbf{T}_{ij}\, \mathbf{m}_{1:t}(i),$$

while it records the maximizing predecessor $\psi_{t+1}(j) = \arg\max_i \mathbf{T}_{ij}\, \mathbf{m}_{1:t}(i)$. With the states in the order (a, b, c),

$$\begin{aligned}
\mathbf{m}_{1:1} &\approx \begin{pmatrix} 0.533 & 0.333 & 0.133 \end{pmatrix}^T, \\
\mathbf{m}_{1:2} &\approx \begin{pmatrix} 0.688 & 0.269 & 0.043 \end{pmatrix}^T, && \psi_2 = (a, b, a), \\
\mathbf{m}_{1:3} &\approx \begin{pmatrix} 0.336 & 0.328 & 0.336 \end{pmatrix}^T, && \psi_3 = (a, b, a), \\
\mathbf{m}_{1:4} &\approx \begin{pmatrix} 0.538 & 0.328 & 0.134 \end{pmatrix}^T, && \psi_4 = (a, b, c).
\end{aligned}$$

The normalization does not change the maximizers. The most likely last coin is $x_4^* = \arg\max_j \mathbf{m}_{1:4}(j) = a$, and backtracking, $x_k^* = \psi_{k+1}(x_{k+1}^*)$, gives $x_3^* = \psi_4(a) = a$, $x_2^* = \psi_3(a) = a$ and $x_1^* = \psi_2(a) = a$. The most likely sequence is (a, a, a, a), with posterior probability ≈ 0.181; the next one is (b, b, b, b), ≈ 0.111. The tie between a and c in $\mathbf{m}_{1:3}$ does not matter, since no path through c at $t = 3$ is the best one into a at $t = 4$.

Smoothing gives the most likely coin at each step, (a, a, b, a), which is a different sequence: the most likely sequence is not the sequence of the most likely states.

Exercise 4 Robot state estimation (Exam, January 2024)

Let us consider a continuous first-order Markov process describing the motion of a robot in a one-dimensional world. On average, the robot moves forward by a distance b at each timestep. The robot's position x_t at time t is not directly observable, but can be estimated using a sensor that measures the position of the robot relative to the sensor. Assuming

- the sensor is fixed at position 0,
- a Gaussian prior $\mathcal{N}(x_0 \mid \mu_0, \sigma_0^2)$ on the initial position of the robot,
- a Gaussian transition model $\mathcal{N}(x_{t+1} \mid x_t + b, \sigma_x^2)$,
- a Gaussian sensor model $\mathcal{N}(e_t \mid x_t, \sigma_e^2)$,

answer the following questions.

(a) Derive the update equations of the Bayes filter. More specifically, since the belief distribution is Gaussian, derive the update equations of its mean μ_{t+1} and variance σ_{t+1}^2.

Solution. This is a Kalman filter with a scalar state x_t and evidence e_t, whose components are $\mathbf{A} = 1$, $\mathbf{b} = b$, $\mathbf{Q} = \sigma_x^2$, $\mathbf{H} = 1$, $\mathbf{d} = 0$ and $\mathbf{R} = \sigma_e^2$; without the drift ($b = 0$), it is the 1d Gaussian random walk. Let $p(x_t \mid e_{1:t}) = \mathcal{N}(x_t \mid \mu_t, \sigma_t^2)$.

Predict step. The second identity of the cheat sheet, with x_t as $\mathbf{x}$ and x_{t+1} as $\mathbf{y}$, gives a joint Gaussian whose marginal is

$$p(x_{t+1} \mid e_{1:t}) = \mathcal{N}(x_{t+1} \mid \mu_t + b, s_{t+1}), \quad s_{t+1} = \sigma_t^2 + \sigma_x^2 :$$

the drift moves the mean and the transition noise adds its variance.

Update step. The same identity, with x_{t+1} as $\mathbf{x}$ and e_{t+1} as $\mathbf{y}$, gives

$$p\left(\begin{pmatrix} x_{t+1} \\ e_{t+1} \end{pmatrix} \middle| e_{1:t}\right) = \mathcal{N}\left(\begin{pmatrix} x_{t+1} \\ e_{t+1} \end{pmatrix} \middle| \begin{pmatrix} \mu_t + b \\ \mu_t + b \end{pmatrix}, \begin{pmatrix} s_{t+1} & s_{t+1} \\ s_{t+1} & s_{t+1} + \sigma_e^2 \end{pmatrix}\right),$$

and the identity for $p(\mathbf{x} \mid \mathbf{y})$ conditions on e_{t+1}. With the Kalman gain

$$k_{t+1} = \frac{s_{t+1}}{s_{t+1} + \sigma_e^2} = \frac{\sigma_t^2 + \sigma_x^2}{\sigma_t^2 + \sigma_x^2 + \sigma_e^2},$$

the update equations are

$$\mu_{t+1} = \mu_t + b + k_{t+1}(e_{t+1} - (\mu_t + b)) = \frac{(\sigma_t^2 + \sigma_x^2)e_{t+1} + \sigma_e^2(\mu_t + b)}{\sigma_t^2 + \sigma_x^2 + \sigma_e^2},$$
$$\sigma_{t+1}^2 = (1 - k_{t+1})(\sigma_t^2 + \sigma_x^2) = \frac{(\sigma_t^2 + \sigma_x^2)\sigma_e^2}{\sigma_t^2 + \sigma_x^2 + \sigma_e^2}.$$

The new mean is a weighted average of the observation e_{t+1} and of the predicted mean $\mu_t + b$.

(b) Do the update equations make sense in the following limiting cases? Use your physical intuition.

- The sensor is extremely noisy.
- The sensor is perfect.
- The transition model is extremely noisy.
- The transition model is perfect.

Solution.

- The sensor is extremely noisy.
  With $\sigma_e^2 \to +\infty$, $k_{t+1} \to 0$ and the update equations become

  $$\mu_{t+1} = \mu_t + b,$$
  $$\sigma_{t+1}^2 = \sigma_t^2 + \sigma_x^2.$$

  As expected, the optimal strategy is to completely ignore the observations and rely solely on the initial belief and the transition model. The uncertainty is indeed the combination of the uncertainty in the transition model and the uncertainty of the previous state, which ultimately originates from the uncertainty in the initial belief.
- The sensor is perfect.
  With $\sigma_e^2 \to 0$, $k_{t+1} \to 1$ and the update equations become

  $$\mu_{t+1} = e_{t+1},$$
  $$\sigma_{t+1}^2 = 0.$$

  As expected, the optimal strategy is to rely solely on the observations. Consequently, no uncertainty arises when estimating the robot's position.
- The transition model is extremely noisy.
  With $\sigma_x^2 \to +\infty$, $k_{t+1} \to 1$ and the update equations become

  $$\mu_{t+1} = e_{t+1},$$
  $$\sigma_{t+1}^2 = \sigma_e^2.$$

  As expected, the optimal strategy is to rely solely on the observations, since nothing reliable can be obtained after applying the transition model. Consequently, the uncertainty in the estimation is that of the sensor model.

- The transition model is perfect.
  With $\sigma_x^2 \to 0$, $k_{t+1} \to \frac{\sigma_t^2}{\sigma_t^2+\sigma_e^2}$ and the update equations become

$$\mu_{t+1} = \mu_t + b + \frac{\sigma_t^2}{\sigma_t^2 + \sigma_e^2}(e_{t+1} - (\mu_t + b)),$$
$$\sigma_{t+1}^2 = \frac{\sigma_t^2 \sigma_e^2}{\sigma_t^2 + \sigma_e^2}.$$

  Although it may seem surprising to still observe the effect of the observations on the estimation, the result is actually perfectly sensible when recalling that an uncertainty σ_0^2 is associated with the first estimate. If the initial position of the robot is known ($\sigma_0^2 \to 0$), then the update equations with the perfect transition model become

$$\mu_{t+1} = \mu_t + b,$$
$$\sigma_{t+1}^2 = 0.$$

  As expected, the optimal strategy is to rely solely on the initial position and the transition model and no uncertainty is associated with the estimation.

(c) Describe one step of a particle filter for this model, from N particles $x_t^1, \ldots, x_t^N$ that represent $p(x_t \mid e_{1:t})$ to particles that represent $p(x_{t+1} \mid e_{1:t+1})$. What do the particles approximate, and when is a particle filter preferable to the update equations of (a)?

Solution.

1. *Propagate:* move each particle through the transition model, $x_{t+1}^i \sim \mathcal{N}(x_t^i + b, \sigma_x^2)$. The particles now represent the predicted belief $p(x_{t+1} \mid e_{1:t})$.
2. *Weight:* give each particle the likelihood of the new evidence, $w^i = \mathcal{N}(e_{t+1} \mid x_{t+1}^i, \sigma_e^2)$.
3. *Resample:* draw N new particles among the x_{t+1}^i, with probabilities proportional to the weights w^i, so that unlikely particles disappear and likely ones are duplicated.

The resampled particles approximate the Gaussian $\mathcal{N}(x_{t+1} \mid \mu_{t+1}, \sigma_{t+1}^2)$ of (a): their mean and variance tend to μ_{t+1} and σ_{t+1}^2 as N grows. Here the update equations are exact and cheaper. A particle filter is preferable when the transition or the sensor model is not linear Gaussian, for instance if the sensor measured the distance $|x_t|$ to the robot rather than its position, since then no closed form exists.

Exercise 5 Ball on a spring

The "Foire de Liège" has a new attraction called "Super Spring Ultra Pro Max XXL", which consists of a ball attached to a spring on a platform. The participants take a seat in the ball, locked at some position. The ball is then released and pulled by the spring, which makes it oscillate back and forth. After a few seconds, the ball is stopped by magnetic brakes. You notice that the final position of the ball is different each time. As the ball is opaque, you wonder if it is possible for the participants to guess where the ball stopped, given their perception of acceleration.

You more or less remember your Newtonian mechanics class and model the movement of the ball as a series of transitions

$$z_t = z_{t-1} + \Delta t\, \dot{z}_{t-1} + \frac{1}{2}\Delta t^2\, \ddot{z}_{t-1}$$
$$\dot{z}_t = \dot{z}_{t-1} + \Delta t\, \ddot{z}_{t-1}$$
$$\ddot{z}_t = \rho w_{t-1} - \kappa\, z_{t-1} - \eta\, \dot{z}_{t-1}$$
$$w_t \mid w_{t-1} \sim \mathcal{N}(\lambda w_{t-1}, \sigma_w^2)$$

where z_t, $\dot{z}_t$ and $\ddot{z}_t$ are respectively the position, velocity and acceleration of the ball at timestep t, w_t is the wind at timestep t, Δt is the time elapsed between $t-1$ and t, ρ is the thrust coefficient of the wind, κ is the stiffness of the spring, η is the friction coefficient of the platform and λ is the persistence of the wind. You estimate that the ball starts at position $z_0 = 10.0 \pm 0.5\,\mathrm{m}$ with a negligible speed $0.0 \pm 0.1\,\mathrm{m\,s^{-1}}$ and acceleration $0.0 \pm 0.1\,\mathrm{m\,s^{-2}}$. You assume that the wind has a stationary distribution. You assume that the human perception of acceleration follows an unbiased Gaussian distribution. You also assume that a good model to describe the uncertainty in your initial guess of the ball's state is a Gaussian model, with the standard deviation given by your estimated bounds.

(a) You wish to estimate the current state of the ball given the perceptions of a participant. Define the components of a Kalman filter in this context.

Solution. The state gathers the variables that evolve over time, $\mathbf{x}_t = \begin{pmatrix} z_t & \dot{z}_t & \ddot{z}_t & w_t \end{pmatrix}^T$, and the evidence e_t is the perceived acceleration. A Kalman filter needs a Gaussian prior $p(\mathbf{x}_0) = \mathcal{N}(\mathbf{x}_0 \mid \boldsymbol{\mu}_0, \boldsymbol{\Sigma}_0)$ and linear Gaussian transition and sensor models

$$\begin{aligned} p(\mathbf{x}_{t+1} \mid \mathbf{x}_t) &= \mathcal{N}(\mathbf{x}_{t+1} \mid \mathbf{A}\mathbf{x}_t + \mathbf{b}, \mathbf{Q}), \\ p(e_t \mid \mathbf{x}_t) &= \mathcal{N}(e_t \mid \mathbf{H}\mathbf{x}_t + \mathbf{d}, \mathbf{R}). \end{aligned}$$

Equivalently, $\mathbf{x}_{t+1} = \mathbf{A}\mathbf{x}_t + \mathbf{b} + \boldsymbol{\epsilon}_t$ and $e_t = \mathbf{H}\mathbf{x}_t + \mathbf{d} + \mathbf{v}_t$, with noises $\boldsymbol{\epsilon}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{Q})$ and $\mathbf{v}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{R})$ independent of $\mathbf{x}_t$: given $\mathbf{x}_t$, the sum $\mathbf{A}\mathbf{x}_t + \mathbf{b} + \boldsymbol{\epsilon}_t$ is Gaussian, with mean $\mathbf{A}\mathbf{x}_t + \mathbf{b}$ and covariance $\mathbf{Q}$. The components are identified in this form.

Transition model. The wind is written in the same form, $w_{t+1} = \lambda w_t + \xi_t$ with $\xi_t \sim \mathcal{N}(0, \sigma_w^2)$, which has the mean and the variance of $\mathcal{N}(\lambda w_t, \sigma_w^2)$. By identification,

$$\mathbf{A} = \begin{pmatrix} 1 & \Delta t & \frac{1}{2}\Delta t^2 & 0 \\ 0 & 1 & \Delta t & 0 \\ -\kappa & -\eta & 0 & \rho \\ 0 & 0 & 0 & \lambda \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \quad \mathbf{Q} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \sigma_w^2 \end{pmatrix}.$$

The first three equations are deterministic: only the wind is noisy.

Sensor model. The participant perceives the acceleration with an unbiased Gaussian noise, whose variance is not given; let us call it σ_e^2. Then

$$\mathbf{H} = \begin{pmatrix} 0 & 0 & 1 & 0 \end{pmatrix}, \quad \mathbf{d} = 0, \quad \mathbf{R} = \sigma_e^2.$$

Prior. From the statement, $z_0 \sim \mathcal{N}(10, 0.5^2)$, $\dot{z}_0 \sim \mathcal{N}(0, 0.1^2)$ and $\ddot{z}_0 \sim \mathcal{N}(0, 0.1^2)$. The distribution of the wind does not change with time, so that

$$\begin{aligned} \mathbb{E}[w_0] = \mathbb{E}[w_1] = \mathbb{E}[\lambda w_0 + \xi_0] &= \lambda \mathbb{E}[w_0], \\ \mathbb{V}[w_0] = \mathbb{V}[w_1] = \mathbb{V}[\lambda w_0 + \xi_0] &= \lambda^2 \mathbb{V}[w_0] + \sigma_w^2, \end{aligned}$$

hence, assuming $|\lambda| < 1$ (otherwise the wind has no stationary distribution), $w_0 \sim \mathcal{N}(0, \frac{\sigma_w^2}{1-\lambda^2})$. The statement mentions no link between the initial uncertainties, so we take them independent; independence implies zero covariance, and $\boldsymbol{\Sigma}_0$ is diagonal:

$$\boldsymbol{\mu}_0 = \begin{pmatrix} 10 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \quad \boldsymbol{\Sigma}_0 = \begin{pmatrix} 0.5^2 & 0 & 0 & 0 \\ 0 & 0.1^2 & 0 & 0 \\ 0 & 0 & 0.1^2 & 0 \\ 0 & 0 & 0 & \frac{\sigma_w^2}{1-\lambda^2} \end{pmatrix}.$$

(b) Express the distribution $p(\mathbf{x}_t \mid e_{1:t})$ with respect to the components defined previously. Use the cheat sheet for Gaussian distributions.

Solution. This is filtering, which alternates a predict step and an update step:

$$p(\mathbf{x}_t \mid e_{1:t}) = \alpha\, p(e_t \mid \mathbf{x}_t) \int p(\mathbf{x}_t \mid \mathbf{x}_{t-1})\, p(\mathbf{x}_{t-1} \mid e_{1:t-1})\, \mathrm{d}\mathbf{x}_{t-1}.$$

The prior is Gaussian and both models are linear Gaussian, so every belief is Gaussian, by induction. Let $p(\mathbf{x}_{t-1} \mid e_{1:t-1}) = \mathcal{N}(\mathbf{x}_{t-1} \mid \boldsymbol{\mu}_{t-1}, \boldsymbol{\Sigma}_{t-1})$.
Predict step. The second identity of the cheat sheet, with $\mathbf{x}_{t-1}$ as $\mathbf{x}$, $\mathbf{x}_t$ as $\mathbf{y}$, $(\mathbf{A}, \mathbf{b}, \mathbf{Q})$ as $(\mathbf{H}, \mathbf{u}, \mathbf{R})$, and everything conditioned on $e_{1:t-1}$, gives the joint distribution

$$p\left(\begin{pmatrix}\mathbf{x}_{t-1}\\ \mathbf{x}_t\end{pmatrix} \middle| e_{1:t-1}\right) = \mathcal{N}\left(\begin{pmatrix}\mathbf{x}_{t-1}\\ \mathbf{x}_t\end{pmatrix} \middle| \begin{pmatrix}\boldsymbol{\mu}_{t-1}\\ \mathbf{A}\boldsymbol{\mu}_{t-1} + \mathbf{b}\end{pmatrix}, \begin{pmatrix}\boldsymbol{\Sigma}_{t-1} & \boldsymbol{\Sigma}_{t-1}\mathbf{A}^T\\ \mathbf{A}\boldsymbol{\Sigma}_{t-1} & \mathbf{A}\boldsymbol{\Sigma}_{t-1}\mathbf{A}^T + \mathbf{Q}\end{pmatrix}\right),$$

whose marginal $p(\mathbf{y})$ is the predicted belief

$$p(\mathbf{x}_t \mid e_{1:t-1}) = \mathcal{N}(\mathbf{x}_t \mid \boldsymbol{\mu}_t^-, \boldsymbol{\Sigma}_t^-), \quad \boldsymbol{\mu}_t^- = \mathbf{A}\boldsymbol{\mu}_{t-1} + \mathbf{b}, \quad \boldsymbol{\Sigma}_t^- = \mathbf{A}\boldsymbol{\Sigma}_{t-1}\mathbf{A}^T + \mathbf{Q}.$$

Update step. The same identity, with $\mathbf{x}_t$ as $\mathbf{x}$, e_t as $\mathbf{y}$ and $(\mathbf{H}, \mathbf{d}, \mathbf{R})$ as $(\mathbf{H}, \mathbf{u}, \mathbf{R})$, gives

$$p\left(\begin{pmatrix}\mathbf{x}_t\\ e_t\end{pmatrix} \middle| e_{1:t-1}\right) = \mathcal{N}\left(\begin{pmatrix}\mathbf{x}_t\\ e_t\end{pmatrix} \middle| \begin{pmatrix}\boldsymbol{\mu}_t^-\\ \mathbf{H}\boldsymbol{\mu}_t^- + \mathbf{d}\end{pmatrix}, \begin{pmatrix}\boldsymbol{\Sigma}_t^- & \boldsymbol{\Sigma}_t^-\mathbf{H}^T\\ \mathbf{H}\boldsymbol{\Sigma}_t^- & \mathbf{H}\boldsymbol{\Sigma}_t^-\mathbf{H}^T + \mathbf{R}\end{pmatrix}\right).$$

Conditioning on e_t with the identity for $p(\mathbf{x} \mid \mathbf{y})$ gives the belief

$$p(\mathbf{x}_t \mid e_{1:t}) = \mathcal{N}(\mathbf{x}_t \mid \boldsymbol{\mu}_t, \boldsymbol{\Sigma}_t),$$

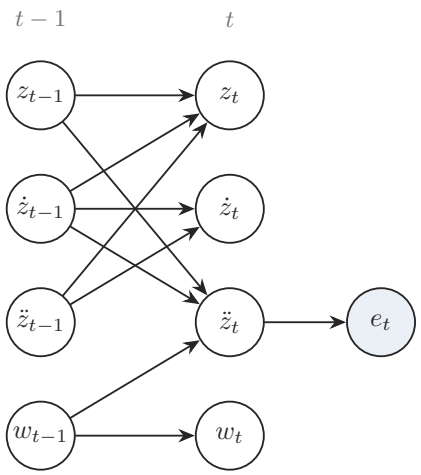
with

$$\begin{aligned}\boldsymbol{\mu}_t &= \boldsymbol{\mu}_t^- + \mathbf{K}_t(e_t - \mathbf{H}\boldsymbol{\mu}_t^- - \mathbf{d}),\\ \boldsymbol{\Sigma}_t &= \boldsymbol{\Sigma}_t^- - \mathbf{K}_t\mathbf{H}\boldsymbol{\Sigma}_t^-,\\ \mathbf{K}_t &= \boldsymbol{\Sigma}_t^-\mathbf{H}^T(\mathbf{H}\boldsymbol{\Sigma}_t^-\mathbf{H}^T + \mathbf{R})^{-1},\end{aligned}$$

where $\mathbf{K}_t$ is the Kalman gain. Here $\mathbf{b} = \mathbf{0}$ and $\mathbf{d} = 0$, and the other components are those of (a).

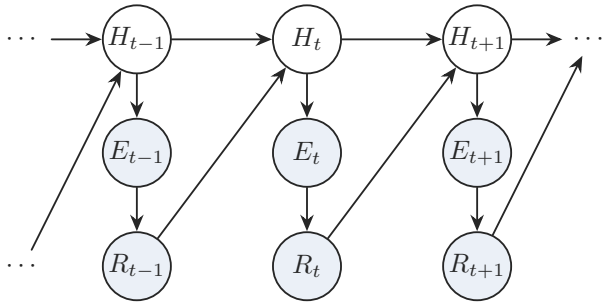
(c) Represent the transition and sensor models as a dynamic Bayesian network.

Solution.



Exercise 6 Human-robot interaction (CS188, Fall 2017)

In the near future, autonomous robots will live among us. Therefore, the robots need to know how to act appropriately in the presence of humans. In this question, we explore a simplified model of this interaction. We assume that we can observe the robot's actions at time t, R_t, and an evidence observation, E_t, directly caused by the (hidden) human state, H_t. Robot actions from the current time-step affect the human state in the next-time step, as illustrated in the Bayesian network given below.



Assuming discrete random variables and given the probability tables $\mathbf{P}(H_0)$, $\mathbf{P}(E_t \mid H_t)$, $\mathbf{P}(R_t \mid E_t)$ and $\mathbf{P}(H_{t+1} \mid H_t, R_t)$, your goal is to derive a procedure to maintain a belief $\mathbf{P}(H_t \mid e_{1:t}, r_{1:t})$ about the state of the human at time t. At each time step, the robot's action r_t is observed after the evidence e_t.

(a) Derive the update step, which incorporates a pair of observations (e_t, r_t) into the predicted belief $\mathbf{P}(H_t \mid e_{1:t-1}, r_{1:t-1})$.

Solution.

$$\begin{aligned}\mathbf{P}(H_t \mid e_{1:t}, r_{1:t}) &= \alpha\, \mathbf{P}(e_t, r_t \mid H_t, e_{1:t-1}, r_{1:t-1})\, \mathbf{P}(H_t \mid e_{1:t-1}, r_{1:t-1}) \\ &= \alpha\, P(r_t \mid e_t)\, \mathbf{P}(e_t \mid H_t)\, \mathbf{P}(H_t \mid e_{1:t-1}, r_{1:t-1}) \\ &= \alpha\, \mathbf{P}(e_t \mid H_t)\, \mathbf{P}(H_t \mid e_{1:t-1}, r_{1:t-1})\end{aligned}$$

The factor $P(r_t \mid e_t)$ does not depend on H_t and is absorbed by the normalization: since R_t depends on E_t only, observing r_t after e_t tells nothing more about H_t, and $\mathbf{P}(H_t \mid e_{1:t}, r_{1:t}) = \mathbf{P}(H_t \mid e_{1:t}, r_{1:t-1})$.

(b) Derive the predict step, which gives the predicted belief $\mathbf{P}(H_{t+1} \mid e_{1:t}, r_{1:t})$ about the state of the human at time $t+1$ from the belief $\mathbf{P}(H_t \mid e_{1:t}, r_{1:t})$.

Solution.

$$\begin{aligned}\mathbf{P}(H_{t+1} \mid e_{1:t}, r_{1:t}) &= \sum_{h_t} \mathbf{P}(H_{t+1}, h_t \mid e_{1:t}, r_{1:t}) \\ &= \sum_{h_t} \mathbf{P}(H_{t+1} \mid h_t, r_t)\, P(h_t \mid e_{1:t}, r_{1:t})\end{aligned}$$

(c) Combine both steps to derive a recursive update of the belief state $\mathbf{P}(H_t \mid e_{1:t}, r_{1:t})$, as observations are collected and time passes.

Solution.

$$\begin{aligned}\mathbf{P}(H_{t+1} \mid e_{1:t+1}, r_{1:t+1}) &= \alpha\, \mathbf{P}(e_{t+1} \mid H_{t+1})\, \mathbf{P}(H_{t+1} \mid e_{1:t}, r_{1:t}) \\ &= \alpha\, \mathbf{P}(e_{t+1} \mid H_{t+1}) \sum_{h_t} \mathbf{P}(H_{t+1} \mid h_t, r_t)\, P(h_t \mid e_{1:t}, r_{1:t})\end{aligned}$$

(d) Let us now assume that all variables are continuous. Discuss how you would compute or approximate the belief state on a computer.

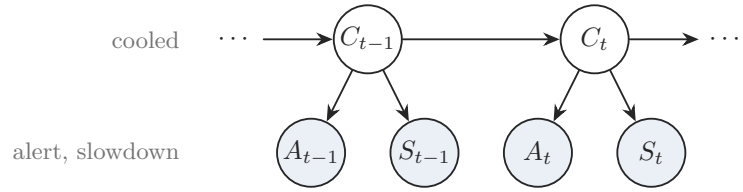
Solution. If the prior is Gaussian and the transition and sensor models are linear Gaussian, the belief stays Gaussian and the Kalman filter updates its mean and covariance in closed form, with the observed action r_t entering the transition model as a known input. Otherwise, the belief has no closed form in general, and a particle filter approximates it with samples: propagate each particle h_t^i through $p(h_{t+1} \mid h_t^i, r_t)$, weight it by $p(e_{t+1} \mid h_{t+1}^i)$, then resample.

Exercise 7 Server cooling (Exam, August 2019; AIMA ch. 14)
A monitoring program wants to know whether a server is cooled enough. Each day, the program checks whether the server raised a temperature alert, and whether it slowed down (throttled) its processors. The program makes the following hypotheses:

- The prior probability that the cooling is enough, with no observations, is 0.7.
- The probability that the cooling is enough on day t is 0.8 given that it was enough on the previous day, and 0.3 otherwise.
- The probability of a temperature alert is 0.2 if the cooling is enough, and 0.7 otherwise.
- The probability of throttling is 0.1 if the cooling is enough, and 0.3 otherwise.

(a) Formulate the environment and hypotheses as a dynamic Bayesian network that the program could use to detect insufficient cooling from a sequence of observations. Provide the associated probability tables.

Solution. Each day t has a hidden state C_t (the cooling is enough) and two evidence variables, A_t (a temperature alert) and S_t (a slowdown), all Boolean:



The prior is $P(c_0) = 0.7$, and

C_{t-1}	$P(c_t \mid C_{t-1})$
true	0.8
false	0.3

C_t	$P(a_t \mid C_t)$
true	0.2
false	0.7

C_t	$P(s_t \mid C_t)$
true	0.1
false	0.3

(b) Reformulate the dynamic Bayesian network as a hidden Markov model that has only a single observation variable. Give the complete probability tables for the model.

Solution. The two evidence variables merge into one, $E_t = (A_t, S_t)$, with four values. Since A_t and S_t are independent given C_t, $P(a, s \mid C_t) = P(a \mid C_t)P(s \mid C_t)$:

C_t	$\neg a, \neg s$	$\neg a, s$	$a, \neg s$	a, s
true	0.72	0.08	0.18	0.02
false	0.21	0.09	0.49	0.21

With the states in the order (true, false) and the evidence values in the order of the table,

$$\mathbf{f}_0 = \begin{pmatrix} 0.7 \\ 0.3 \end{pmatrix}, \quad \mathbf{T} = \begin{pmatrix} 0.8 & 0.2 \\ 0.3 & 0.7 \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} 0.72 & 0.08 & 0.18 & 0.02 \\ 0.21 & 0.09 & 0.49 & 0.21 \end{pmatrix}.$$

(c) For the sequence $e_{1:3}$ of observations "no alert, no throttling", "alert, no throttling" and "alert, throttling", calculate the distributions $\mathbf{P}(C_t \mid e_{1:t})$ and $\mathbf{P}(C_t \mid e_{1:3})$ for $t \in \{1, 2, 3\}$.

Solution. The observations are $e_1 = (\neg a, \neg s)$, $e_2 = (a, \neg s)$ and $e_3 = (a, s)$, so that $\mathbf{O}_1 = \mathrm{diag}(0.72, 0.21)$, $\mathbf{O}_2 = \mathrm{diag}(0.18, 0.49)$ and $\mathbf{O}_3 = \mathrm{diag}(0.02, 0.21)$. Filtering, $\mathbf{f}_{1:t} = \alpha\, \mathbf{O}_t \mathbf{T}^T \mathbf{f}_{1:t-1}$, gives

$$\mathbf{f}_{1:1} = \alpha\, \mathbf{O}_1 \begin{pmatrix} 0.65 & 0.35 \end{pmatrix}^T \approx \begin{pmatrix} 0.864 & 0.136 \end{pmatrix}^T,$$
$$\mathbf{f}_{1:2} = \alpha\, \mathbf{O}_2 \begin{pmatrix} 0.732 & 0.268 \end{pmatrix}^T \approx \begin{pmatrix} 0.501 & 0.499 \end{pmatrix}^T,$$
$$\mathbf{f}_{1:3} = \alpha\, \mathbf{O}_3 \begin{pmatrix} 0.551 & 0.449 \end{pmatrix}^T \approx \begin{pmatrix} 0.104 & 0.896 \end{pmatrix}^T,$$

where each vector is the prediction $\mathbf{T}^T \mathbf{f}_{1:t-1}$. Smoothing, $\mathbf{P}(C_k \mid e_{1:3}) = \alpha\, \mathbf{f}_{1:k} \times \mathbf{b}_{k+1:3}$ with $\mathbf{b}_{k+1:3} = \mathbf{T}\mathbf{O}_{k+1}\mathbf{b}_{k+2:3}$ and $\mathbf{b}_{4:3} = \begin{pmatrix} 1 & 1 \end{pmatrix}^T$, uses

$$\mathbf{b}_{3:3} = \begin{pmatrix} 0.058 \\ 0.153 \end{pmatrix}, \quad \mathbf{b}_{2:3} \approx \begin{pmatrix} 0.0233 \\ 0.0556 \end{pmatrix},$$

and gives

$$\mathbf{P}(C_1 \mid e_{1:3}) \approx \begin{pmatrix} 0.728 & 0.272 \end{pmatrix}^T,$$
$$\mathbf{P}(C_2 \mid e_{1:3}) \approx \begin{pmatrix} 0.276 & 0.724 \end{pmatrix}^T,$$
$$\mathbf{P}(C_3 \mid e_{1:3}) = \mathbf{f}_{1:3} \approx \begin{pmatrix} 0.104 & 0.896 \end{pmatrix}^T.$$

The alerts of days 2 and 3 make insufficient cooling likely, and lower in hindsight the probability that the cooling was enough on day 1 (from 0.864 to 0.728) and on day 2 (from 0.501 to 0.276).

Quiz

1. The normal distribution $\mathcal{N}(\mu, \sigma^2)$ is described by the density function
   - ○ $p(x) = \frac{1}{\sigma} \exp\left(-(z + \exp(-z))\right)$, with $z = \frac{x-\mu}{\sigma}$.
   - ○ $p(x) = \frac{1}{2\sigma} \exp\left(-\frac{|x-\mu|}{\sigma}\right)$.
   - ● $p(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$.
   - ○ $p(x) = \dfrac{1}{\pi\sigma\left[1 + \left(\frac{x-\mu}{\sigma}\right)^2\right]}$.
2. The Kalman filter requires the specification of ...
   - ○ a prior $p(\mathbf{x}_0)$ and a transition model $p(\mathbf{x}_{t+1} \mid \mathbf{x}_t)$.
   - ○ a transition model $p(\mathbf{x}_{t+1} \mid \mathbf{x}_t)$ and a sensor model $p(\mathbf{e}_t \mid \mathbf{x}_t)$.
   - ● a prior $p(\mathbf{x}_0)$, a transition model $p(\mathbf{x}_{t+1} \mid \mathbf{x}_t)$, and a sensor model $p(\mathbf{e}_t \mid \mathbf{x}_t)$.
   - ○ a prior $p(\mathbf{x}_0)$, a prior $p(\mathbf{e}_0)$, a transition model $p(\mathbf{x}_{t+1} \mid \mathbf{x}_t)$, and a sensor model $p(\mathbf{e}_t \mid \mathbf{x}_t)$.
3. In filtering, the belief state is updated at each timestep using ...
   - ○ the current belief and the transition model.
   - ○ the current belief, the sensor model and the new evidence.
   - ○ the transition model, the sensor model and the new evidence.
   - ● the current belief, the transition model, the sensor model and the new evidence.

4. Let us assume a Markov process with hidden state variables $\mathbf{X}_t$, evidence variables $\mathbf{E}_t$, transition model $\mathbf{P}(\mathbf{X}_{t+1} \mid \mathbf{X}_t)$ and sensor model $\mathbf{P}(\mathbf{E}_t \mid \mathbf{X}_t)$. Which of the following is true?

   ◯ The Bayes filter estimates $\mathbf{P}(\mathbf{X}_{1:t} \mid \mathbf{e}_{1:t})$.

   ◯ Smoothing consists of computing $\mathbf{P}(\mathbf{X}_{t+1} \mid \mathbf{e}_{1:t+1})$ by pushing the belief state $\mathbf{P}(\mathbf{X}_t \mid \mathbf{e}_{1:t})$ forward through the transition model and then updating this new belief state with the evidence $\mathbf{e}_{t+1}$.

   ◯ The Kalman filter computes the belief state exactly for non-linear transition models.

   ◉ The Viterbi algorithm computes the most likely sequence of state values $\mathbf{x}_{1:t}$ given the evidence $\mathbf{e}_{1:t}$.